

Agricultural Productivity Analysis Dashboard

Total Rainfall

Total Rainfall
877.68K

Average Rice Yield

Avg Rice Yield
1.61K

Average Wheat Yield

Avg Wheat Yield
1.45K

Average Kharif Yield

Avg Kharif Yield
441.96

States

Total States
18

Years Covered

Total Years
52

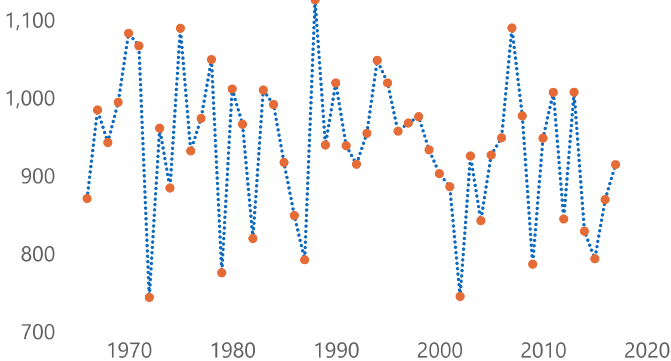
Year

All

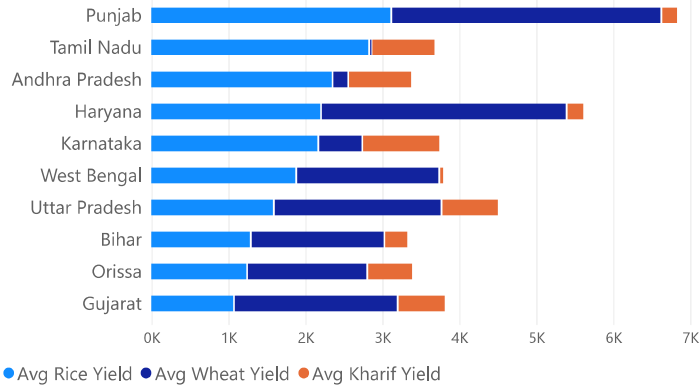
Region

All

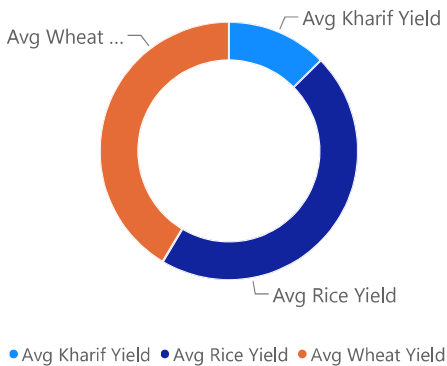
Average Rainfall by Years



Top 10 States by Average Crop Yield (Crop wise)



Crop Yield Distribution by Crop Type



Page navigator (Press Ctrl + Left Click on page name on mouse for direct navigation)

Agricultural Productivity
Analysis Dashboard

Rainfall Trend Over Time vs
Regions

Avg. Rainfall vs Avg. Rice Yield

Avg. Rainfall vs Avg. Wheat
Yield

Avg. Rainfall vs Avg. Kharif
Yield

Regional Agricultural
Productivity Analysis

Yield Decomposition Analysis

Insights and
Recommendations

Temporal trends of rainfall across different regions.

Year

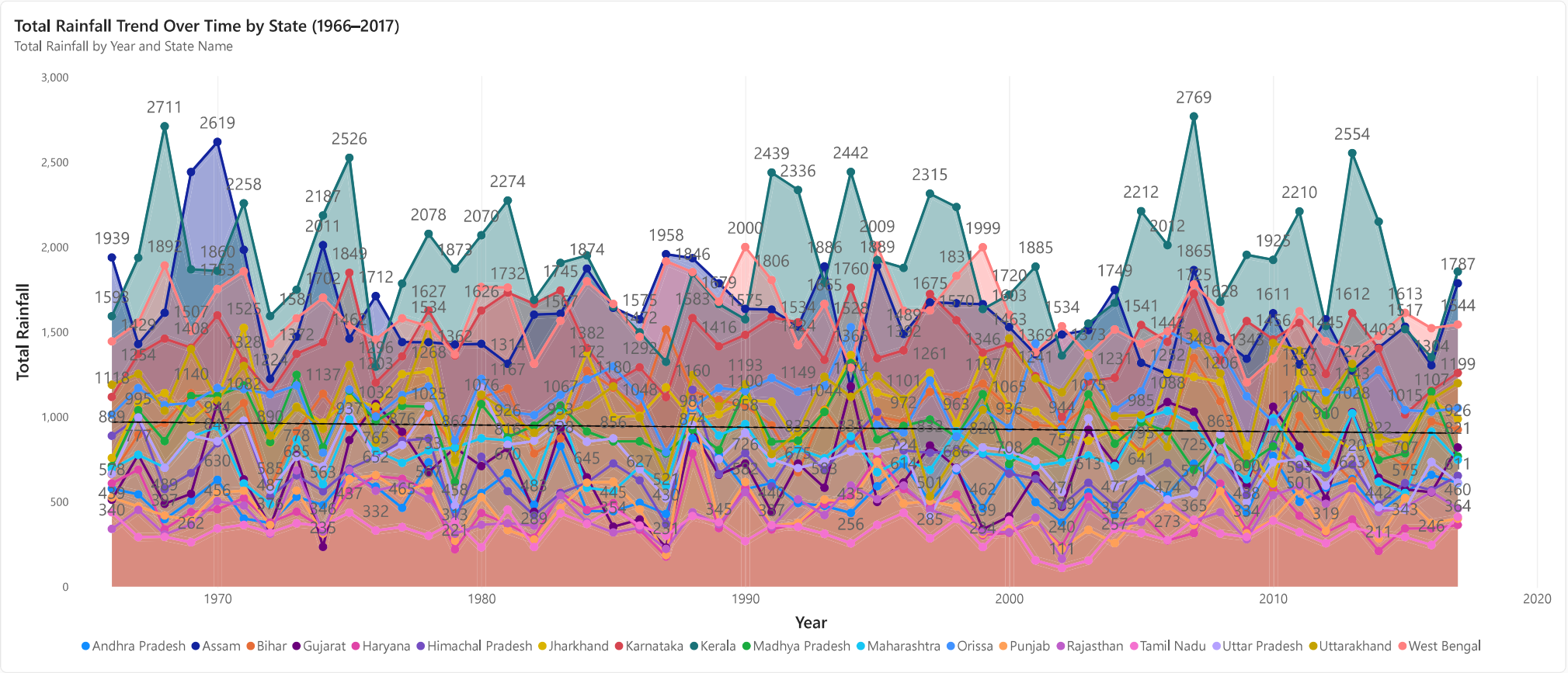
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Region

All

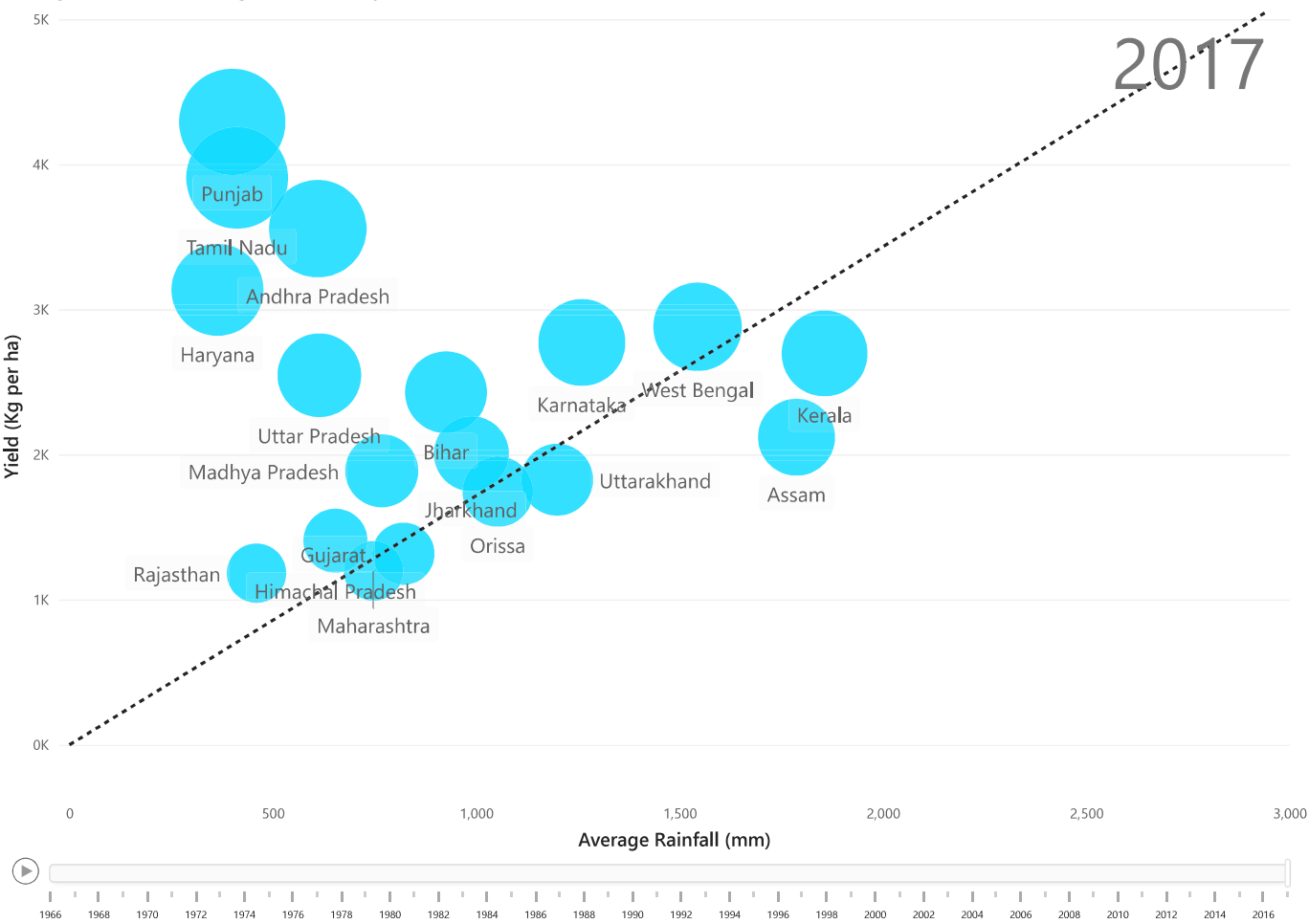
Rain Category

All

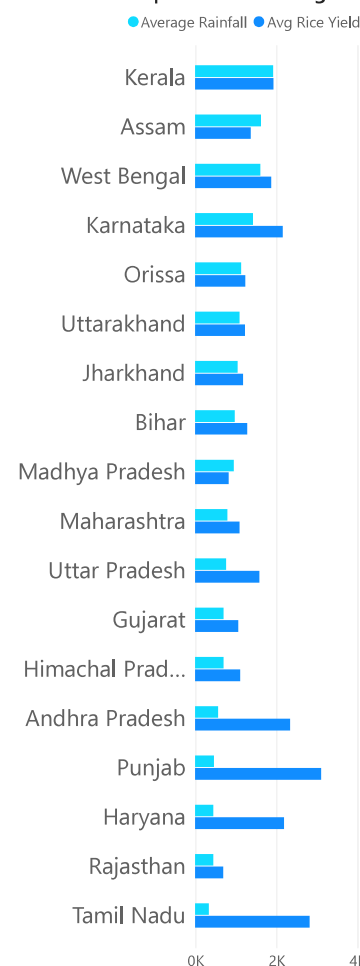


Correlation between rainfall volumes and crop yields (Rice)

Average Rainfall vs Average Rice Yield by State



State-wise Comparison — Average ...



Year

All

Region

All

Rain Category

All

Yield Summary

Average Rainfall

937.69

Max Rice Yield

4.66K

Avg Rice Yield

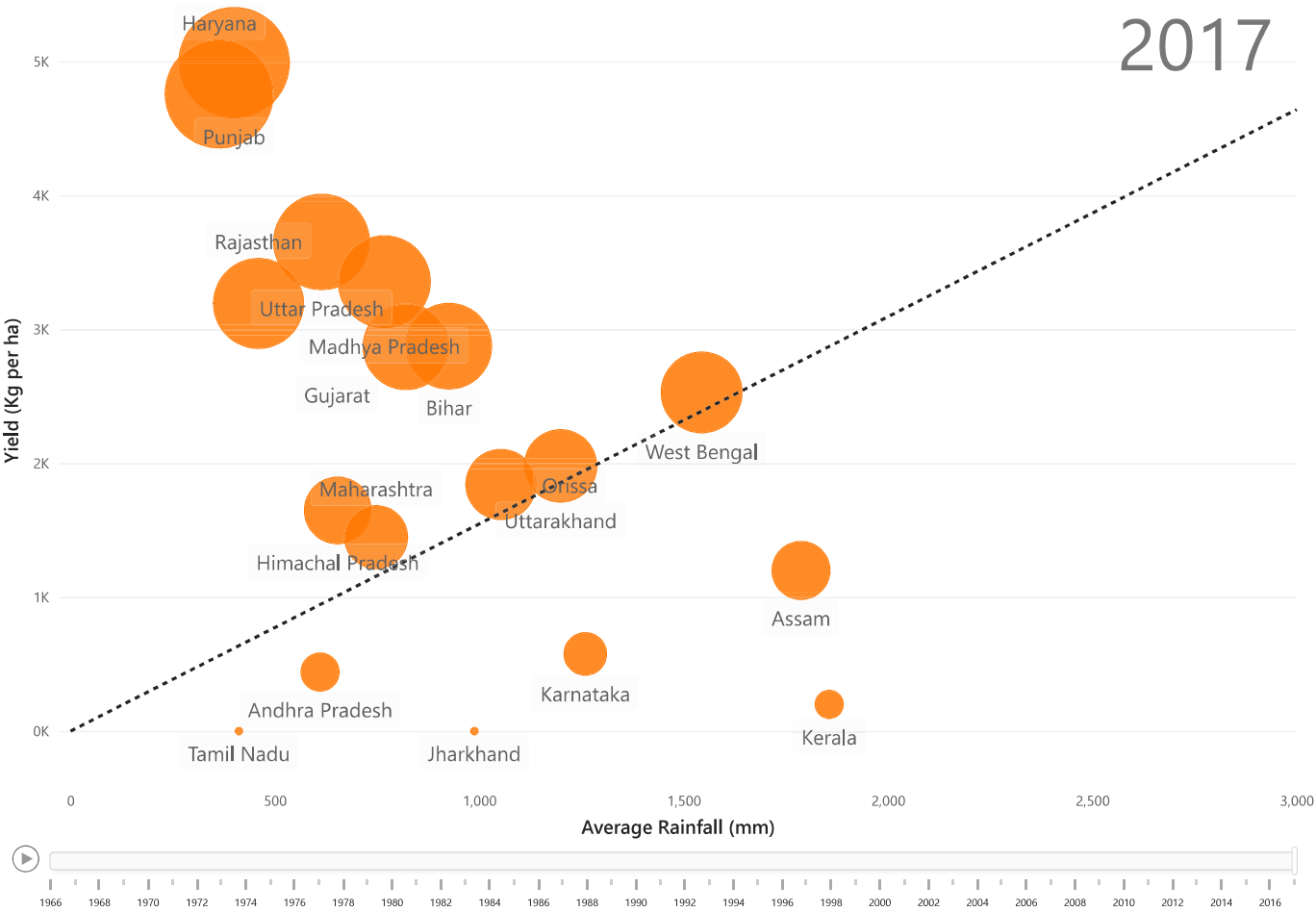
1.61K

Min Rice Yield

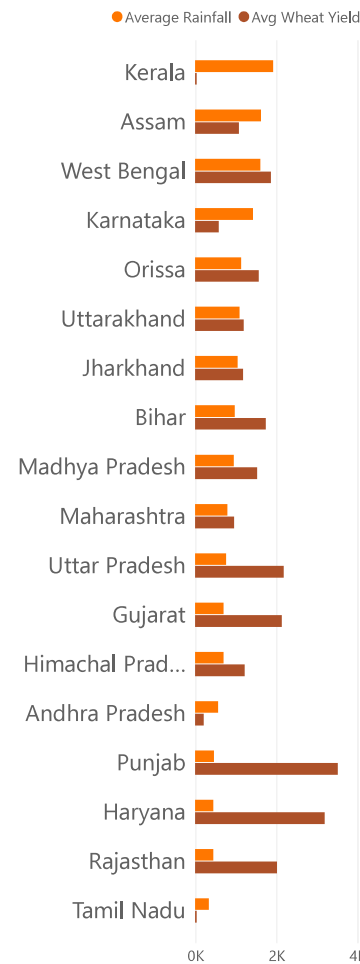
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Correlation between rainfall volumes and crop yields (Wheat)

Average Rainfall vs Average Wheat Yield by State



State-wise Comparison — Average ...



Year

All

Region

All

Rain Category

All

Yield Summary

Average Rainfall

937.69

Max Wheat Yield

5.09K

Avg Wheat Yield

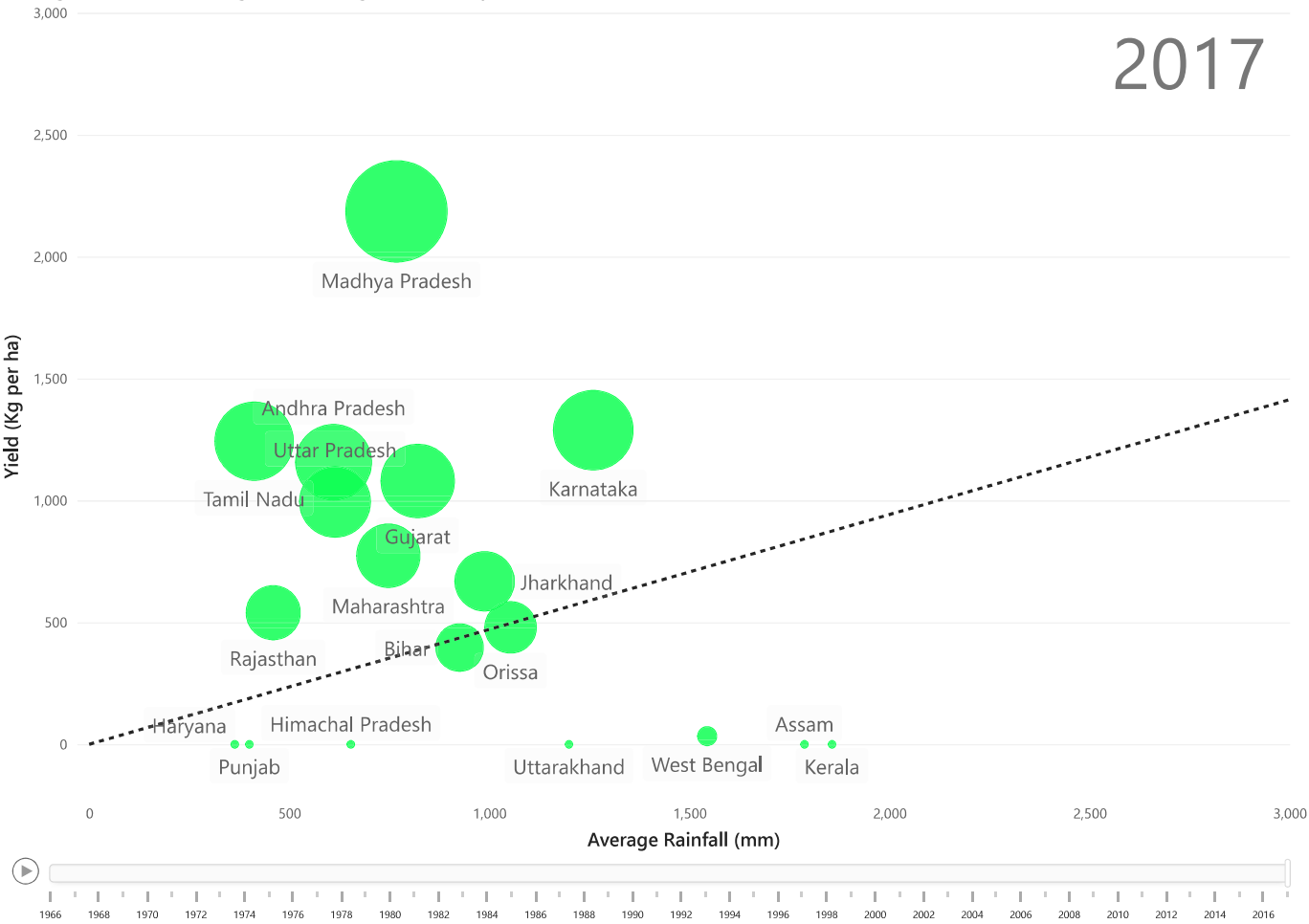
1.45K

Min Wheat Yield

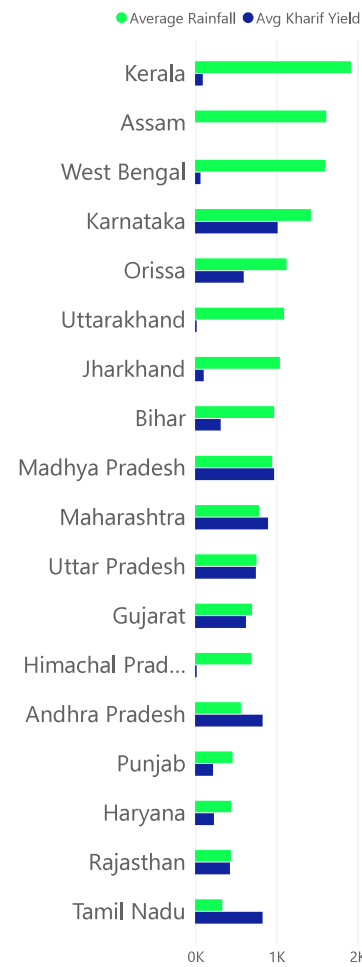
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Correlation between rainfall volumes and crop yields (Kharif)

Average Rainfall vs Average Kharif Sorghum Yield by State



State-wise Comparison — Average ...



Year

All

Region

All

Rain Category

All

Yield Summary

Average Rainfall

937.69

Max Kharif Yield

2.64K

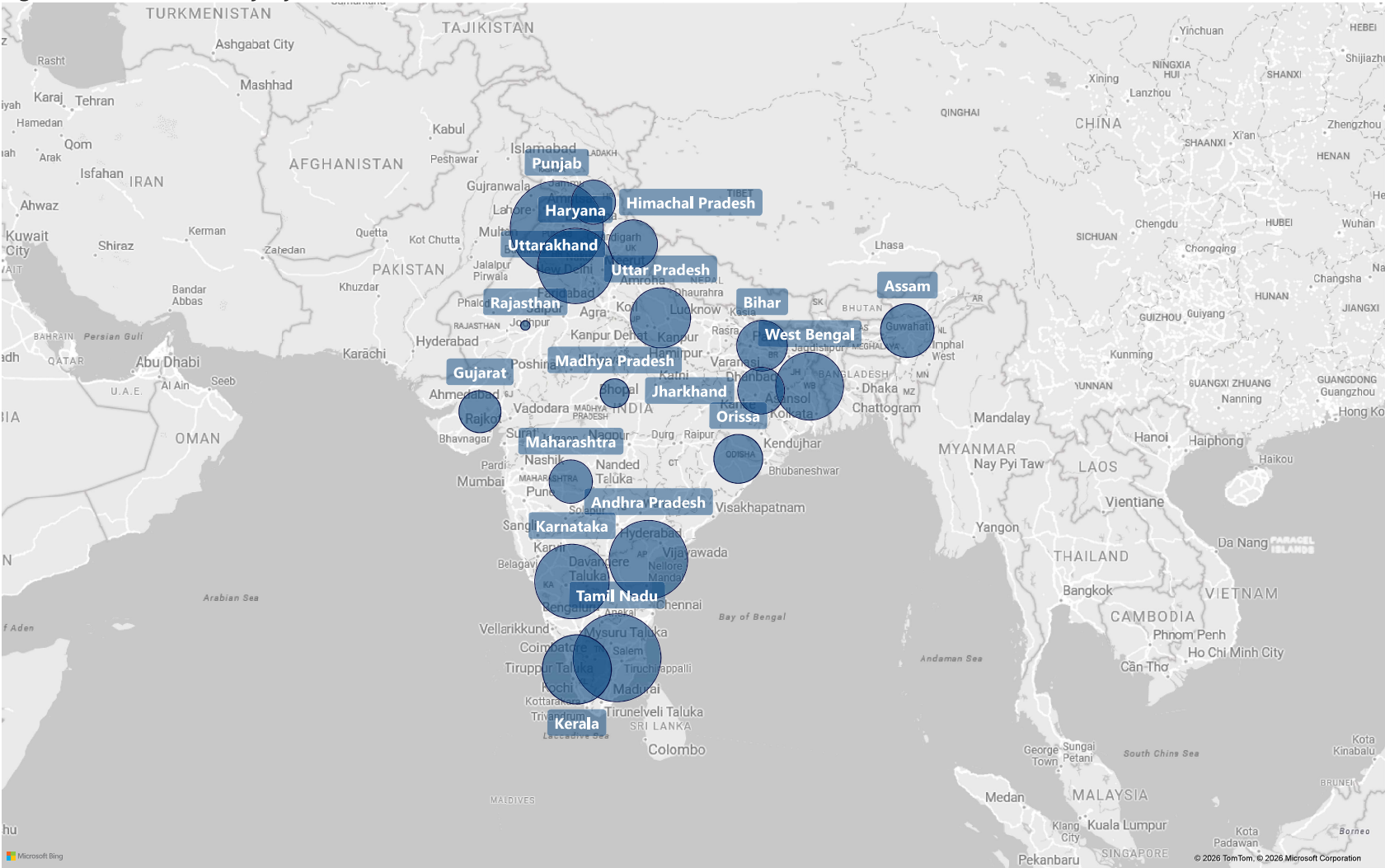
Avg Kharif Yield

441.96

Min Kharif Yield

0.00

Agricultural Productivity by State - Rainfall & Yield



Year

All

Region

All

Rain Category

All

Crop Type

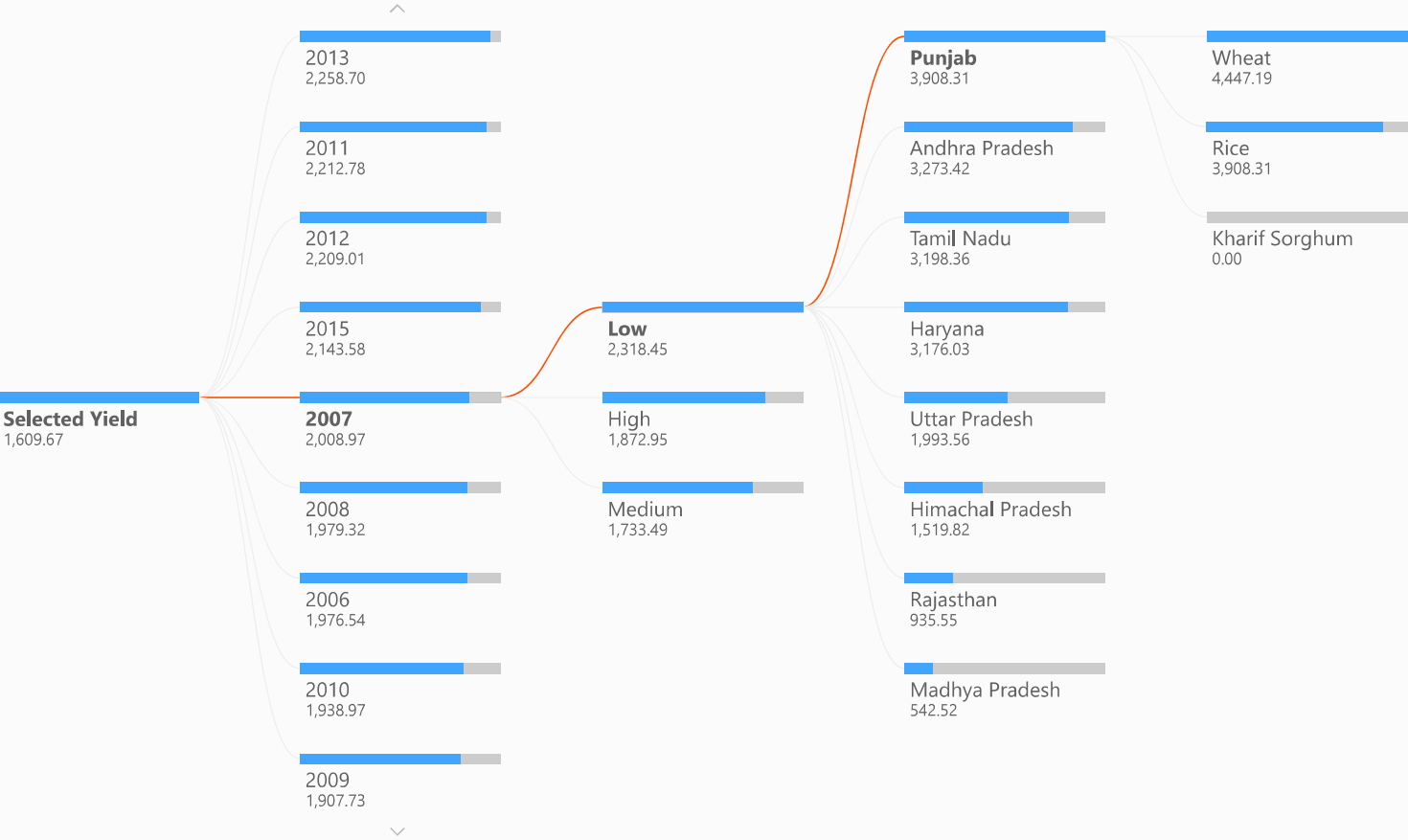
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Key Metrics Summary

Average Rainfall (mm)	937.69
Sum of RICE PRODUCTION (1000 t...	243.35K
Sum of WHEAT PRODUCTION (100...	176.85K
Sum of KHARIF SORGHUM PROD...	11.98K

Drill-Down: Yield by State & Crop

Year	Rainfall Category	State Name	Crop
2007	Low	Punjab	



Year

All

Region

All

Rain Category

All

Key Metrics Summary

Total Rainfall (mm)

877.68K

Average Rainfall (mm)

937.69

Avg Kharif Yield (Kg per ha)

441.96

Avg Rice Yield (Kg per ha)

1.61K

Avg Wheat Yield (Kg per ha)

1.45K

Key Insights from the Dashboard : After analyzing the dashboard, I identified the following key insights.

1. Rainfall Varies Significantly Across Years : The rainfall trend analysis shows noticeable fluctuations in rainfall over different years. Some years experienced comparatively higher rainfall, while others recorded lower values. These variations indicate that rainfall patterns are not consistent and can directly influence agricultural production.

2. Rice Yield is Closely Related to Rainfall : From the rainfall versus rice yield analysis, I observed that rice production generally performs better in areas receiving adequate rainfall. Since rice is a water-intensive crop, sufficient rainfall contributes positively to its productivity. However, extremely high rainfall does not always result in proportionally higher yields, indicating that other factors also influence production.

The scatter plot for Rice shows a moderate positive correlation between rainfall and yield across states.

Highest Avg Rainfall Kerala (~4,000+ mm) and Assam (~3,000 mm) receive the most rainfall.

Top Rice Yielders Punjab (~4.66K kg/ha) achieves maximum rice yield despite moderate rainfall.

Punjab's high rice yield despite moderate rainfall (~1,000 mm) suggests irrigation infrastructure plays a critical role beyond natural rainfall.

3. Wheat Yield is Less Dependent on Rainfall : Compared to rice, wheat yield showed a weaker relationship with rainfall. This suggests that wheat production depends not only on rainfall but also on irrigation facilities, farming techniques, and other agricultural practices.

Wheat shows an inverse relationship with extreme rainfall — the highest wheat yields occur in relatively drier, well-irrigated states.

Max Wheat Yield 5.09K kg/ha — Punjab leads, followed by Haryana and Uttar Pradesh.

Rainfall-Yield Pattern Wheat yields are HIGH in low-to-moderate rainfall zones (500–1,500 mm).

High-Rainfall States Kerala and Assam, despite heavy rainfall, show comparatively low wheat yields.

Best Performers Punjab and Haryana are above the trend line — they over-perform relative to rainfall.

Wheat is a Rabi (winter) crop — it thrives in cooler, drier conditions. Excess monsoon rainfall in states like Kerala is counterproductive for wheat cultivation.

4. Kharif Crops Depend Strongly on Monsoon Rainfall : The analysis of Kharif crops indicates that their productivity is strongly influenced by monsoon rainfall. Regions receiving stable rainfall generally achieve better Kharif yields, highlighting the importance of timely and adequate